

Omprakash Sahani

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Education

Sanjay Ghodawat University <i>B.Tech in Computer Science and Engineering (CGPA 8.40/10.0, First Class with Distinction)</i> Relevant coursework: Machine Learning, Data Structures & Algorithms, Operating Systems, Computer Networks, Distributed Systems, Distributed & Parallel Computing, Database Management Systems.	2020 – 2023 Kolhapur, MH, India
Dr. D. Y. Patil Polytechnic <i>Diploma in Computer Engineering (MSBTE, First Class)</i>	2017 – 2020 Kolhapur, MH, India

Experience

LeRobot State Atlas – Dream Machines Collaboration <i>Open-Source Collaborator</i> - Built LeRobot State Atlas to analyze dual-arm LeRobot datasets using URDF-based forward kinematics, tool trajectories, and voxelized workspace coverage; validated a 100-episode, 47,455-frame pilot and cut local export time by 12.4% while cutting peak RSS by ~29.7 MiB. - Added a Gaussian Splat environment workflow using Nerfstudio and the Spark Gaussian renderer; evaluated a 466k-Gaussian reconstruction and a 250k-splat (~3.8 MiB) browser representation that rendered at ~46 FPS in local diagnostics.	Jul 2026 – Present Remote
Independent ML Systems Engineering <i>Open-Source Projects</i> Build open-source ML systems for evaluation, benchmarking, distributed training, search, and observability.	Jul 2024 – Present Remote

Projects

SplatLab (In Progress) GitHub Gaussian Splatting tool for asset inspection, camera analysis, rendering, and performance diagnostics.
SearchEval Lab (In Progress) GitHub Search evaluation framework for TF-IDF, BM25, hybrid retrieval, ranking metrics, weak-query analysis, and regression checks.
Atlas AI GitHub ML systems platform for transformers, distributed runtime, inference serving, benchmarking, and observability.
Codex Benchmark Guardian GitHub Benchmark regression and repair workflow for pull requests with deterministic analysis, FastAPI/Next.js tooling, protected verification, and human approval.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL.
ML & Systems: Transformers, Distributed Training, Inference Serving, Gaussian Splatting, Robotics Kinematics.
Evaluation & Performance: Retrieval & Evaluation, Benchmarking, Regression Analysis, Distributed Runtime Profiling.
Frameworks & Libraries: FastAPI, Next.js, React, Three.js, LeRobot, NumPy, scikit-learn, Nerfstudio, Spark (Gaussian Splatting), Pytest.
Infrastructure & Tools: Git/GitHub, GitHub Actions, Linux/WSL, REST APIs, GitHub Apps & Webhooks, SQLite, Vercel.

Additional Information

OpenAI Build Week 2026 – Codex Benchmark Guardian GitHub Live Demo Submitted a bounded Codex repair workflow for benchmark regressions with protected verification and human approval.
micro1 Frontier Engineering Challenge 2026 – EvidencePatch GitHub Selected as a Top 50 Applicant ; built a governed evidence-to-code workflow with deterministic provenance verification.
FortyGuard Hackathon '26 – ThermalShift AI GitHub Live Demo Built a thermal-aware GPU scheduler using FortyGuard data and OR-Tools CP-SAT; reduced modeled thermal exposure by 20.1% vs. First Available in the summer historical replay while preserving 100% deadline satisfaction .
Leadership: Led a 3-member B.Tech ML project (2022) using machine learning models for diabetes prediction and a CNN for brain-tumor image classification (A+ grade); led a 5-member team to 3rd place in the Best Out of Waste competition (2019).
Problem Solving: Solved 200+ Data Structures & Algorithms problems on LeetCode .
Languages: English (Fluent), Hindi (Native), Marathi (Native), Marwari (Native), Sanskrit (Beginner).